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In [ ]: PAWAR SAKSHI MAHENDRA  
        ROLL NO - 49  
        PRACTICAL NO - 04
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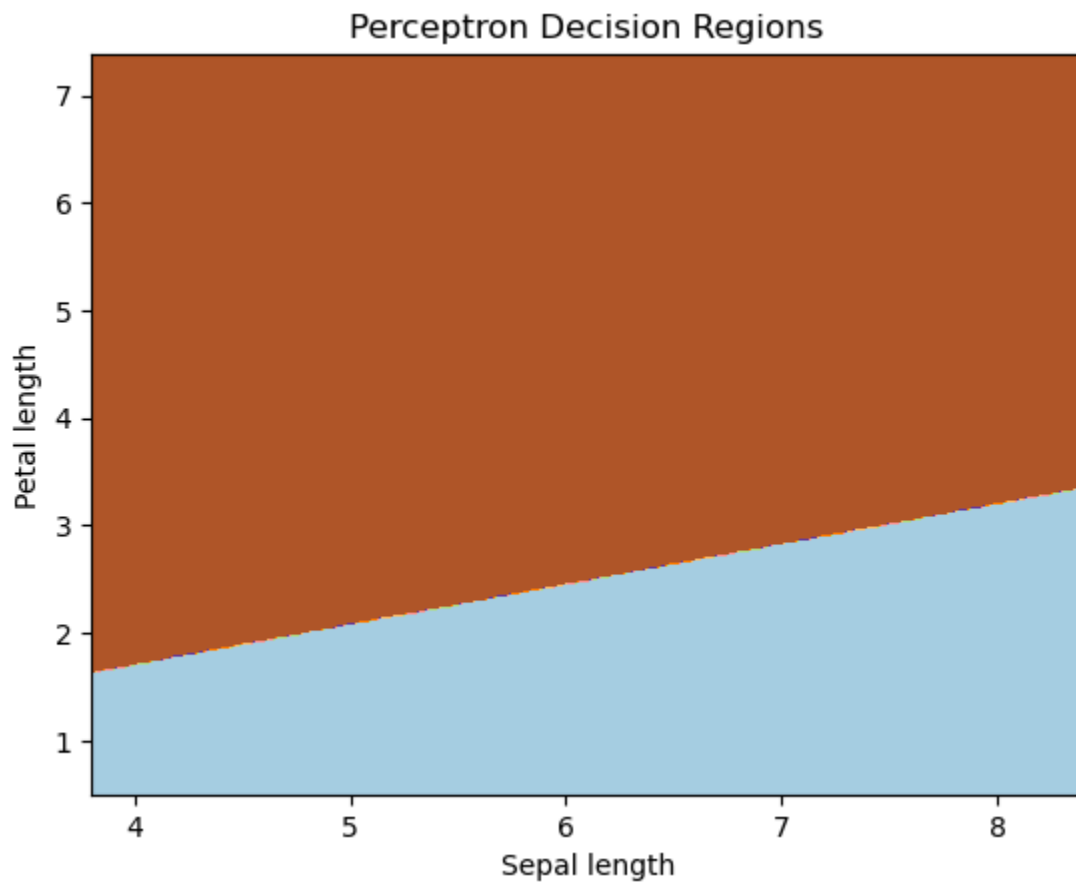
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In [1]: import numpy as np  
import matplotlib.pyplot as plt  
from sklearn.datasets import load_iris  
  
# load iris dataset  
iris = load_iris()  
  
# extract sepal length and petal length  
X = iris.data[:, [0, 2]]  
y = iris.target  
  
# convert to binary classification  
y = np.where(y == 0, 0, 1) # setosa = 0, others = 1  
  
# initialize weights and bias  
w = np.zeros(2)  
b = 0  
  
# learning rate and epochs  
lr = 0.1  
epochs = 50  
  
# perceptron function  
def perceptron(x, w, b):  
    z = np.dot(x, w) + b  
    return np.where(z >= 0, 1, 0)  
  
# training  
for epoch in range(epochs):  
    for i in range(len(X)):  
        x = X[i]  
        target = y[i]  
  
        output = perceptron(x, w, b)  
        error = target - output  
  
        w += lr * error * x  
        b += lr * error  
  
# decision boundary plot  
x_min, x_max = X[:, 0].min() - 0.5, X[:, 0].max() + 0.5  
y_min, y_max = X[:, 1].min() - 0.5, X[:, 1].max() + 0.5  
  
xx, yy = np.meshgrid(np.arange(x_min, x_max, 0.02),  
                     np.arange(y_min, y_max, 0.02))  
  
Z = perceptron(np.c_[xx.ravel(), yy.ravel()], w, b)  
Z = Z.reshape(xx.shape)
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plt.contourf(xx, yy, Z, cmap=plt.cm.Paired)

# plot data points
plt.scatter(X[:, 0], X[:, 1], c=y, cmap=plt.cm.Paired)

plt.xlabel('Sepal length')
plt.ylabel('Petal length')
plt.title('Perceptron Decision Regions')

plt.show()
```



In []: